

Data Visualisation for Economics

Analysis income taxes by wikipedia scrapping

Lucas Simatovic

Master 2 : Economics DADEE track

Assignment 3

décembre 08, 2025

Abstract

Through this exercise, we are going to analyse the income highest tax by countries in the world thanks to a wikipedia table. We start by cleaning first with a cleaning function without characters not relevant, then construct a continent column which associate countries to the continent, different visualisations like bar charts or world map, comparing VAT to highest income tax, handle missing data and finally a free visualisation which tell us a story from INSEE.

Libraries

```
library(rvest)
library(magrittr)
library(tidyverse)
library(stringr)
library(dplyr)
library(countrycode)
```

First part

```
url <- "https://en.wikipedia.org/wiki/List_of_countries_by_tax_rates"

tax_data_raw <- read_html(url) %>%
  html_element("table.wikitable") %>%
  html_table(fill = TRUE)

tax_data_raw
```

```
## # A tibble: 233 x 10
##   'Tax jurisdiction' Corporate 'Individual income' 'Individual income'
##   <chr>               <chr>               <chr>               <chr>
## 1 Tax jurisdiction    Corporate          Lowest             Highest
## 2 Afghanistan        20%[2]            0%[3]              20%[3]
## 3 Albania             15%[6]            0%[7]              23%[7]
## 4 Algeria             19-26%[9]         0%[10]             35%[10]
## 5 American Samoa      34%[6]            4%[12] [13]        6%[13]
```

```
## 6 Andorra 10%[15] 0%[16] 10%[16]
## 7 Angola 30%[18] 0%[18] 17%[18]
## 8 Anguilla 0%[20] 0%[20] 0%[20]
## 9 Antigua and Barbuda 25%[21] 0%[22] 0%[22]
## 10 Argentina 35% (residents)1~ 9%[24] 35%[24]
## # i 223 more rows
## # i 6 more variables: 'Capital gains[1]' <chr>, Wealth <chr>, Property <chr>,
## # 'Inheritance / Estate' <chr>, 'VAT or GSTor Sales' <chr>,
## # 'Further reading' <chr>
```

```
tax_data <- tax_data_raw %>%
  select(
    Country = 1,
    CorporateTax = contains("Corporate"),
    IncomeTaxTop = contains("Individual"),
    VAT = contains("VAT")
  ) %>%
  select(1:5)
```

tax_data

```
## # A tibble: 233 x 5
##   Country CorporateTax IncomeTaxTop1 IncomeTaxTop2 VAT
##   <chr> <chr> <chr> <chr> <chr>
## 1 Tax jurisdiction Corporate Lowest Highest VAT ~
## 2 Afghanistan 20%[2] 0%[3] 20%[3] 0% (~
## 3 Albania 15%[6] 0%[7] 23%[7] 20% ~
## 4 Algeria 19-26%[9] 0%[10] 35%[10] 19% ~
## 5 American Samoa 34%[6] 4%[12] [13] 6%[13] 0%[1~
## 6 Andorra 10%[15] 0%[16] 10%[16] 4.5%~
## 7 Angola 30%[18] 0%[18] 17%[18] 14%[~
## 8 Anguilla 0%[20] 0%[20] 0%[20] 0%[2~
## 9 Antigua and Barbuda 25%[21] 0%[22] 0%[22] 15%[~
## 10 Argentina 35% (residents)15% (no~ 9%[24] 35%[24] 21%[~
## # i 223 more rows
```

```
tax_data$IncomeTaxTop1 <- NULL
tax_data <- tax_data[-1, ]
colnames(tax_data) <- c(
  "Country", "Corporate_Tax", "IncomeTaxTop2" = "Top_Marginal_Income_Tax", "VAT")
```

tax_data

```
## # A tibble: 232 x 4
##   Country Corporate_Tax Top_Marginal_Income_~1 VAT
##   <chr> <chr> <chr> <chr>
## 1 Afghanistan 20%[2] 20%[3] 0% (~
## 2 Albania 15%[6] 23%[7] 20% ~
## 3 Algeria 19-26%[9] 35%[10] 19% ~
## 4 American Samoa 34%[6] 6%[13] 0%[1~
## 5 Andorra 10%[15] 10%[16] 4.5%~
## 6 Angola 30%[18] 17%[18] 14%[~
## 7 Anguilla 0%[20] 0%[20] 0%[2~
```

```
## 8 Antigua and Barbuda 25%[21] 0%[22] 15%[~
## 9 Argentina 35% (residents)15% (non-res~ 35%[24] 21%[~
## 10 Armenia 18%[25] 22%[25] 20%[~
## # i 222 more rows
## # i abbreviated name: 1: Top_Marginal_Income_Tax
```

Cleaning function

```
clean_rate <- function(x) {
  x <- str_remove_all(x, "\\[.*?\\]|%|\\*")
  sapply(x, function(cell_content) {
    nums_str <- unlist(str_extract_all(cell_content, "[0-9.]+"))
    nums <- suppressWarnings(parse_number(nums_str))
    nums <- nums[!is.na(nums)]
    nums <- nums[nums < 100]
    if (length(nums) == 0) return(NA_real_)
    max(nums, na.rm = TRUE)
  })
}
```

```
tax_data_clean <- tax_data %>%
  mutate(
    Country = str_remove_all(Country, "\\[.*?\\]|%|\\*") %>% str_trim(),
    IncomeTaxTop = clean_rate(Top_Marginal_Income_Tax),
    CorporateTax = clean_rate(Corporate_Tax),
    VAT = clean_rate(VAT)
  ) %>%
  filter(!is.na(Country) & Country != "")

tax_data_clean$Top_Marginal_Income_Tax <- NULL
tax_data_clean$Corporate_Tax <- NULL

tax_data_clean
```

```
## # A tibble: 232 x 4
##   Country VAT IncomeTaxTop CorporateTax
##   <chr> <dbl> <dbl> <dbl>
## 1 Afghanistan 0 20 20
## 2 Albania 20 23 15
## 3 Algeria 19 35 26
## 4 American Samoa 0 6 34
## 5 Andorra 9.5 10 10
## 6 Angola 14 17 30
## 7 Anguilla 0 0 0
## 8 Antigua and Barbuda 15 0 25
## 9 Argentina 21 35 35
## 10 Armenia 20 22 18
## # i 222 more rows
```

```
write.csv(tax_data_clean, "tax_data_clean.csv")
```

Manual implementation of data

This new dataset is the same as the professor by implementing data manually from my previous scrape. It's not use for next visualizations. There is no big differences between `tax_data_clean` and `tax_data_clean_new`. Maybe the previous scrapping was made few years ago and now data differ.

```
tax_data_clean_new <- tax_data_clean %>%
  mutate(
    CorporateTax = case_when(
      Country == "Austria" ~ 25.0,
      Country == "Azores" ~ 13.6,
      Country == "Barbados" ~ 5.5,
      Country == "Belize" ~ 0.75,
      Country == "Brunei" ~ 1.0,
      Country == "Canada" ~ 26.5,
      Country == "Czech Republic" ~ 19.0,
      Country == "Denmark" ~ 22.0,
      Country == "Estonia" ~ 0.0,
      Country == "Finland" ~ 20.0,
      Country == "Georgia" ~ 0.0,
      Country == "Latvia" ~ 0.0,
      Country == "Madeira" ~ 5.0,
      Country == "Marshall Islands" ~ 0.8,
      Country == "Monaco" ~ 0.0,
      Country == "Norway" ~ 23.0,
      Country == "Pakistan" ~ 0.0,
      Country == "Portugal" ~ 21.0,
      Country == "Saudi Arabia" ~ 2.5,
      Country == "Somalia" ~ 0.0,
      Country == "Sudan" ~ 5.0,
      Country == "Tajikistan" ~ 13.0,
      Country == "United Arab Emirates" ~ 0.0,
      Country == "United Kingdom" ~ 19.0,
      Country == "Uzbekistan" ~ 12.0,
      Country == "U.S. Virgin Islands" ~ 21.0,
      TRUE ~ CorporateTax
    ),
    IncomeTaxTop = case_when(
      Country == "Bahamas" ~ 0.0,
      Country == "Chile" ~ 35.0,
      Country == "Colombia" ~ 35.0,
      Country == "Egypt" ~ 22.5,
      Country == "France" ~ 47.2,
      Country == "Guyana" ~ 40.0,
      Country == "Hungary" ~ 15.0,
      Country == "India" ~ 30.0,
      Country == "Ireland" ~ 47.2,
      Country == "Jamaica" ~ 25.0,
      Country == "Kazakhstan" ~ 10.0,
```

```

Country == "Kenya" ~ 30.0,
Country == "Maldives" ~ 0.0,
Country == "Mauritius" ~ 20.0,
Country == "Moldova" ~ 12.0,
Country == "Montenegro" ~ 12.65,
Country == "Norway" ~ 40.0,
Country == "Pakistan" ~ 35.0,
Country == "Poland" ~ 41.0,
Country == "Russia" ~ 15.0,
Country == "Somalia" ~ 0.0,
Country == "South Korea" ~ 42.0,
Country == "Tajikistan" ~ 13.0,
Country == "Ukraine" ~ 18.0,
Country == "United Kingdom" ~ 62.0,
Country == "United States" ~ 45.38,
TRUE ~ IncomeTaxTop
),
VAT = case_when(
  Country == "Andorra" ~ 4.5,
  Country == "Angola" ~ 10.0,
  Country == "Azores" ~ 4.0,
  Country == "Brazil" ~ 20.0,
  Country == "Brunei" ~ 55.0,
  Country == "Canada" ~ 5.0,
  Country == "Curaçao" ~ 6.0,
  Country == "Ecuador" ~ 12.0,
  Country == "Greece" ~ 24.0,
  Country == "India" ~ 28.0,
  Country == "Iraq" ~ 0.0,
  Country == "Israel" ~ 17.0,
  str_detect(Country, "Kosovo") ~ 18.0,
  Country == "North Korea" ~ 4.0,
  Country == "Madeira" ~ 5.0,
  Country == "Monaco" ~ 19.6,
  Country == "Montenegro" ~ 21.0,
  Country == "Pakistan" ~ 17.0,
  Country == "Panama" ~ 7.0,
  Country == "San Marino" ~ 0.0,
  Country == "Slovakia" ~ 20.0,
  TRUE ~ VAT
)
)
tax_data_clean_new

```

```

## # A tibble: 232 x 4
##   Country          VAT IncomeTaxTop CorporateTax
##   <chr>          <dbl>         <dbl>         <dbl>
## 1 Afghanistan      0             20             20
## 2 Albania           20             23             15
## 3 Algeria           19             35             26
## 4 American Samoa    0              6             34
## 5 Andorra           4.5            10             10
## 6 Angola            10             17             30

```

```
## 7 Anguilla 0 0 0
## 8 Antigua and Barbuda 15 0 25
## 9 Argentina 21 35 35
## 10 Armenia 20 22 18
## # i 222 more rows
```

```
write.csv(tax_data_clean_new, "tax_data_clean_new.csv")
```

Micronesia is not defined in the wikipedia table libraries. So I have implemented it associated with the continent Oceania. And other countries too to match with my scrape.

```
df_analysis <- tax_data_clean %>%
  mutate(Continent = countrycode(sourcevar = Country,
                                origin = "country.name",
                                destination = "continent")) %>%
  mutate(Continent = case_when(
    Country == "Azores" ~ "Europe",
    str_detect(Country, "Kosovo") ~ "Europe",
    Country == "Madeira" ~ "Europe",
    #####
    str_detect(Country, "Micronesia") ~ "Oceania",
    #####
    Country == "Saint Martin" ~ "Americas",
    Country == "Sark" ~ "Europe",
    TRUE ~ Continent
  )) %>%
  filter(!is.na(Continent))

df_analysis %>%
  filter(
    Country %in% c("Azores", "Madeira", "Saint Martin", "Sark") |
    str_detect(Country, "Kosovo") |
    str_detect(Country, "Micronesia")
  ) %>%
  select(Country, Continent, everything())
```

```
## # A tibble: 6 x 5
##   Country      Continent  VAT IncomeTaxTop CorporateTax
##   <chr>        <chr>    <dbl>      <dbl>      <dbl>
## 1 Azores      Europe     16      36.2      16.8
## 2 Micronesia  Oceania    NA       NA       21
## 3 Kosovo      Europe      0       10       10
## 4 Madeira     Europe     22      45.1      20
## 5 Saint Martin Americas    NA       NA       20
## 6 Sark        Europe      0        0        0
```

```
summary_stats <- df_analysis %>%
  group_by(Continent) %>%
  summarise(
    Average_Income_Tax_Top = mean(IncomeTaxTop, na.rm = TRUE),
    Average_VAT = mean(VAT, na.rm = TRUE)
```

```
)

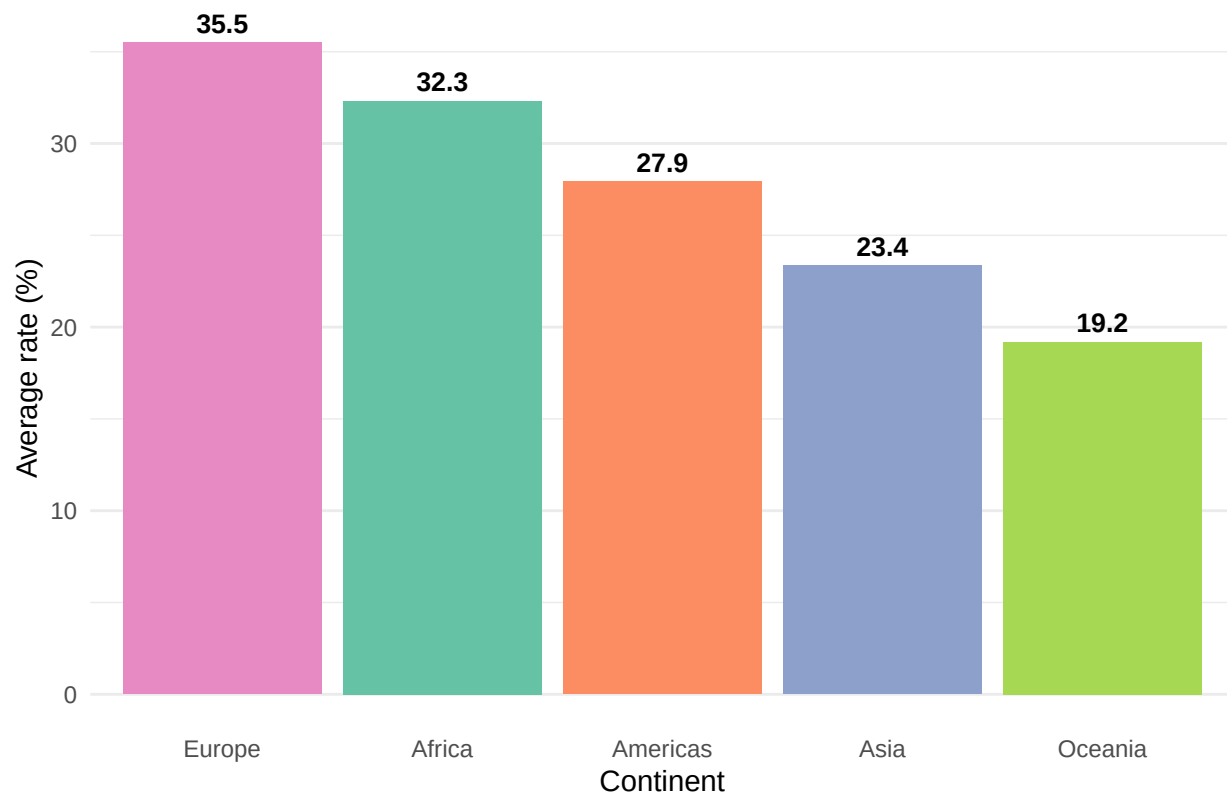
print(summary_stats)

## # A tibble: 5 x 3
##   Continent Average_Income_Tax_Top Average_VAT
##   <chr>           <dbl>         <dbl>
## 1 Africa             32.3           15.2
## 2 Americas            27.9           12.4
## 3 Asia               23.4           11.6
## 4 Europe             35.5           17.4
## 5 Oceania            19.2            6.55
```

The continent which has the highest average income tax rate is Europe. The continent which has the second highest average income tax rate is Africa. The continent which has the highest average VAT rate is Europe too.

```
ggplot(summary_stats,
  aes(x = reorder(Continent, -Average_Income_Tax_Top),
    y = Average_Income_Tax_Top,
    fill = Continent)) +
  geom_col(show.legend = FALSE) +
  scale_fill_brewer(palette = "Set2") +
  geom_text(aes(label = round(Average_Income_Tax_Top, 1)),
    vjust = -0.5, size = 3.5, fontface = "bold") +
  labs(title = "Average Income tax rate by continent",
    x = "Continent",
    y = "Average rate (%)") +
  theme_minimal() +
  theme(panel.grid.major.x = element_blank())
)
```

Average Income tax rate by continent



```
summary_long <- summary_stats %>%
  pivot_longer(cols = c(Average_Income_Tax_Top, Average_VAT),
               names_to = "Tax_Type",
               values_to = "Rate")
```

```
summary_long
```

```
## # A tibble: 10 x 3
##   Continent Tax_Type      Rate
##   <chr>      <chr>      <dbl>
## 1 Africa    Average_Income_Tax_Top 32.3
## 2 Africa    Average_VAT            15.2
## 3 Americas  Average_Income_Tax_Top 27.9
## 4 Americas  Average_VAT            12.4
## 5 Asia      Average_Income_Tax_Top 23.4
## 6 Asia      Average_VAT            11.6
## 7 Europe    Average_Income_Tax_Top 35.5
## 8 Europe    Average_VAT            17.4
## 9 Oceania   Average_Income_Tax_Top 19.2
## 10 Oceania   Average_VAT             6.55
```

```
ggplot(summary_long, aes(x = Continent, y = Rate, fill = Tax_Type)) +
  geom_col(position = position_dodge(width = 0.8), width = 0.7) +
  geom_text(aes(label = round(Rate, 1)),
            position = position_dodge(width = 0.8),
```



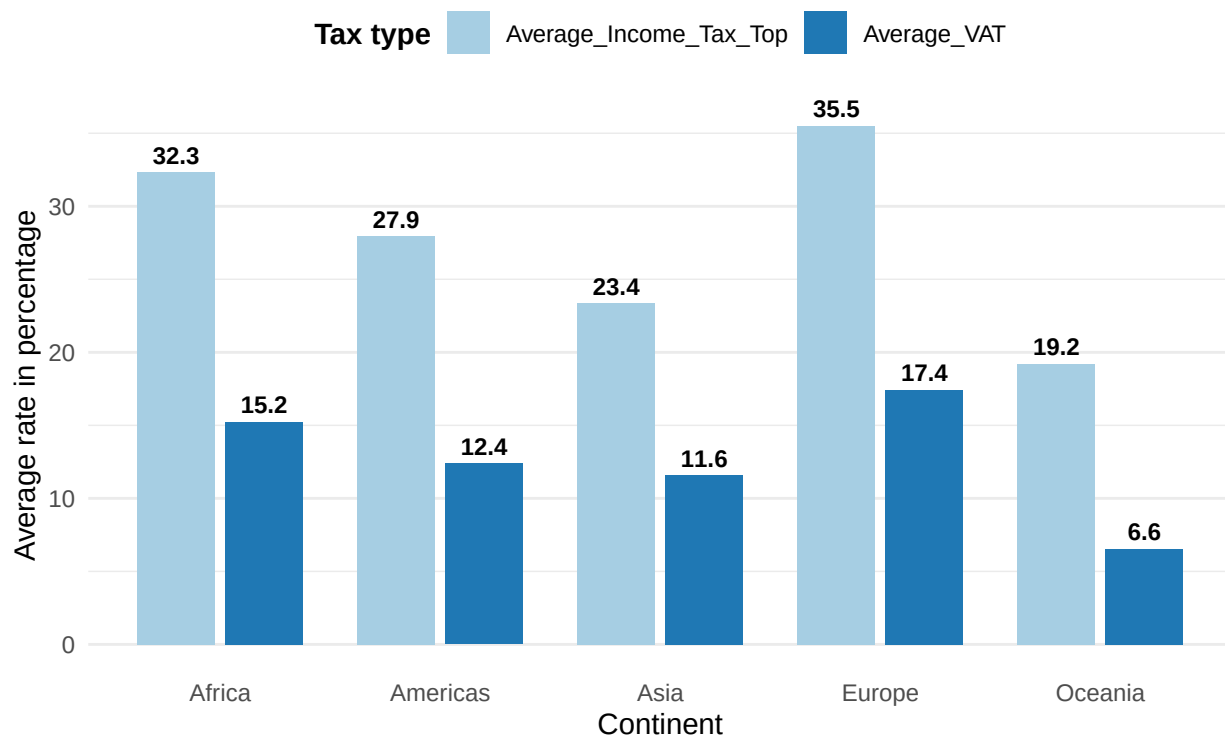
```

    vjust = -0.5, size = 3, fontface = "bold") +
  labs(title = "VAT vs Income tax rate",
        subtitle = "Average rate by continent",
        x = "Continent",
        y = "Average rate in percentage",
        fill = "Tax type") +
  scale_fill_brewer(palette = "Paired") +
  theme_minimal() +
  theme(
    panel.grid.major.x = element_blank(),
    legend.position = "top",
    legend.title = element_text(face = "bold")
  )

```

VAT vs Income tax rate

Average rate by continent



```

world_map <- map_data("world")

df_analysis$iso3 <- countrycode(df_analysis$Country, "country.name", "iso3c")

df_analysis

```

```

## # A tibble: 232 x 6
##   Country          VAT IncomeTaxTop CorporateTax Continent iso3
##   <chr>          <dbl>         <dbl>         <dbl> <chr>    <chr>
## 1 Afghanistan      0             20             20 Asia    AFG
## 2 Albania           20             23             15 Europe  ALB

```

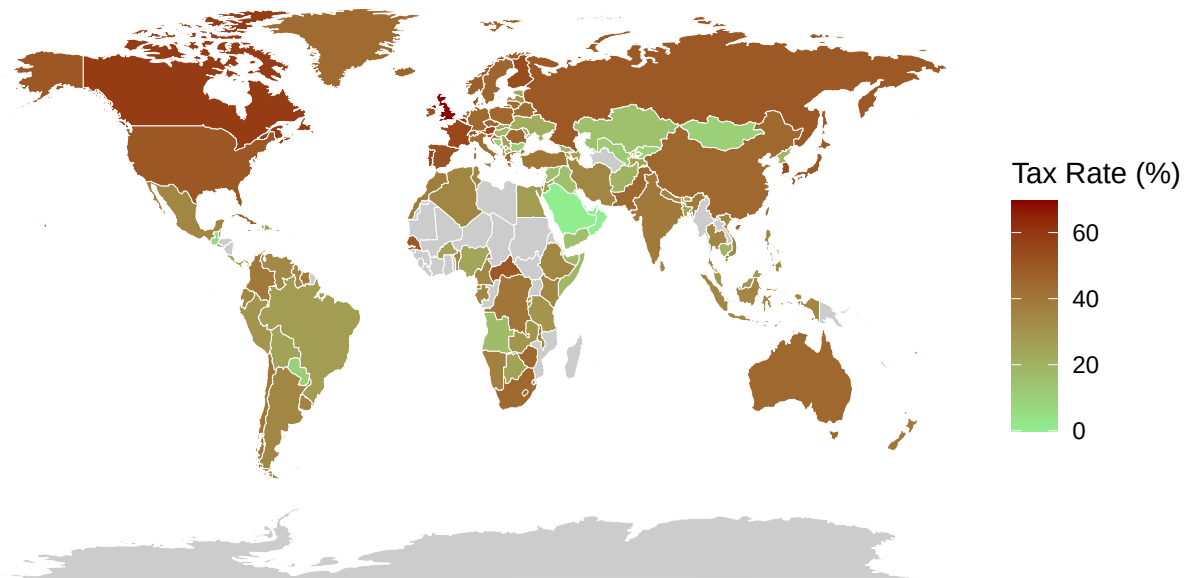
```
## 3 Algeria          19          35          26 Africa    DZA
## 4 American Samoa   0           6          34 Oceania   ASM
## 5 Andorra          9.5         10          10 Europe    AND
## 6 Angola           14          17          30 Africa    AGO
## 7 Anguilla         0           0           0 Americas  AIA
## 8 Antigua and Barbuda 15         0          25 Americas  ATG
## 9 Argentina        21          35          35 Americas  ARG
## 10 Armenia         20          22          18 Asia      ARM
## # i 222 more rows
```

```
world_map$iso3 <- countrycode(world_map$region, "country.name", "iso3c")

map_data_joined <- left_join(world_map, df_analysis, by = "iso3")

ggplot(map_data_joined, aes(long, lat, group = group, fill = IncomeTaxTop)) +
  geom_polygon(color = "white", size = 0.1) +
  scale_fill_gradient(low = "lightgreen", high = "darkred", na.value = "gray80") +
  labs(title = "Top Marginal Income Tax Rate by Country",
       fill = "Tax Rate (%)") +
  theme_void() +
  coord_fixed(1.3)
```

Top Marginal Income Tax Rate by Country



```
df_imputed <- df_analysis %>%
  mutate(IncomeTax_Imputed = ifelse(is.na(IncomeTaxTop), CorporateTax, IncomeTaxTop))
```

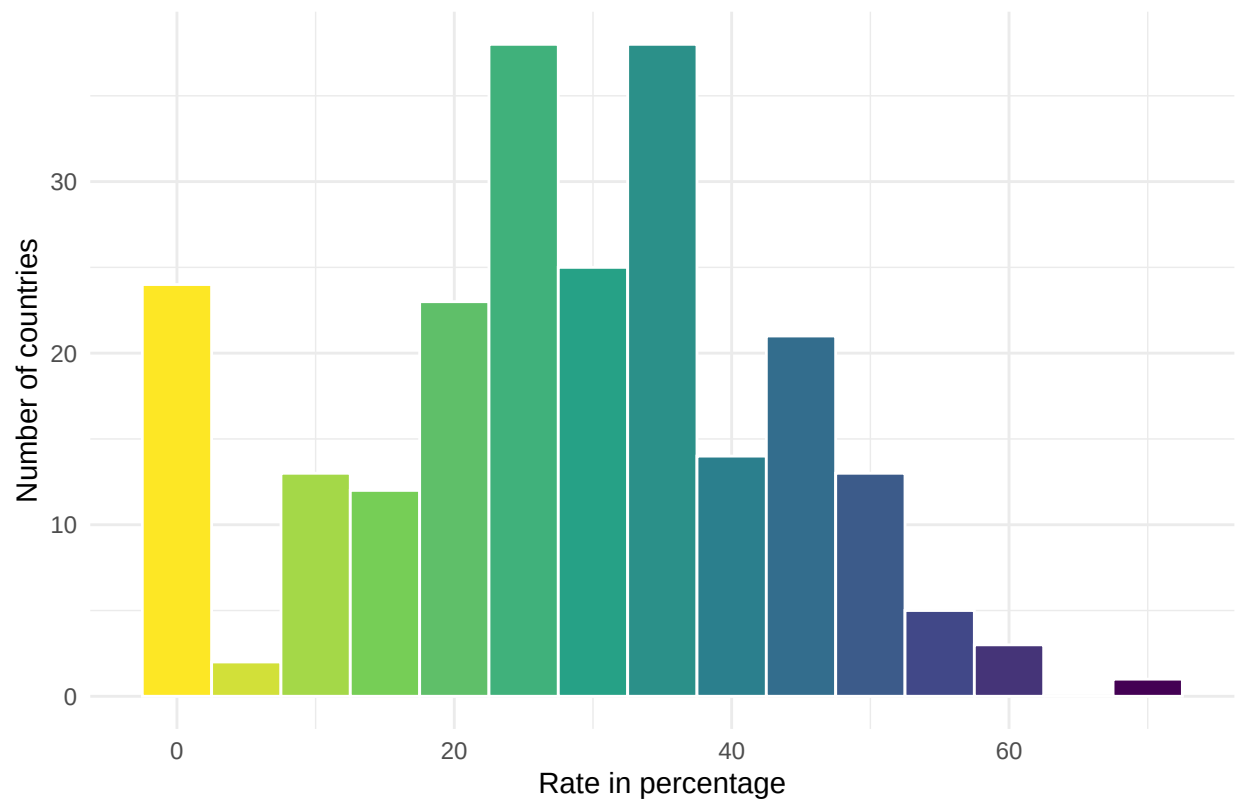
```
df_imputed$CorporateTax <- NULL
df_imputed$VAT <- NULL
```

```
df_imputed
```

```
## # A tibble: 232 x 5
##   Country      IncomeTaxTop Continent iso3 IncomeTax_Imputed
##   <chr>          <dbl> <chr>    <chr>          <dbl>
## 1 Afghanistan      20 Asia      AFG             20
## 2 Albania           23 Europe    ALB             23
## 3 Algeria           35 Africa    DZA             35
## 4 American Samoa     6 Oceania   ASM              6
## 5 Andorra           10 Europe    AND             10
## 6 Angola            17 Africa    AGO             17
## 7 Anguilla           0 Americas AIA              0
## 8 Antigua and Barbuda 0 Americas ATG              0
## 9 Argentina          35 Americas ARG             35
## 10 Armenia           22 Asia      ARM             22
## # i 222 more rows
```

```
ggplot(df_imputed, aes(x = IncomeTax_Imputed)) +
  geom_histogram(aes(fill = after_stat(x)), binwidth = 5, color = "white") +
  scale_fill_viridis_c(option = "viridis", direction = -1, name = "Rate in percentage") +
  labs(title = "Distribution of tax rates after imputation of corporate taxes",
       x = "Rate in percentage", y = "Number of countries") +
  theme_minimal() +
  theme(legend.position = "none")
```

Distribution of tax rates after imputation of corporate taxes

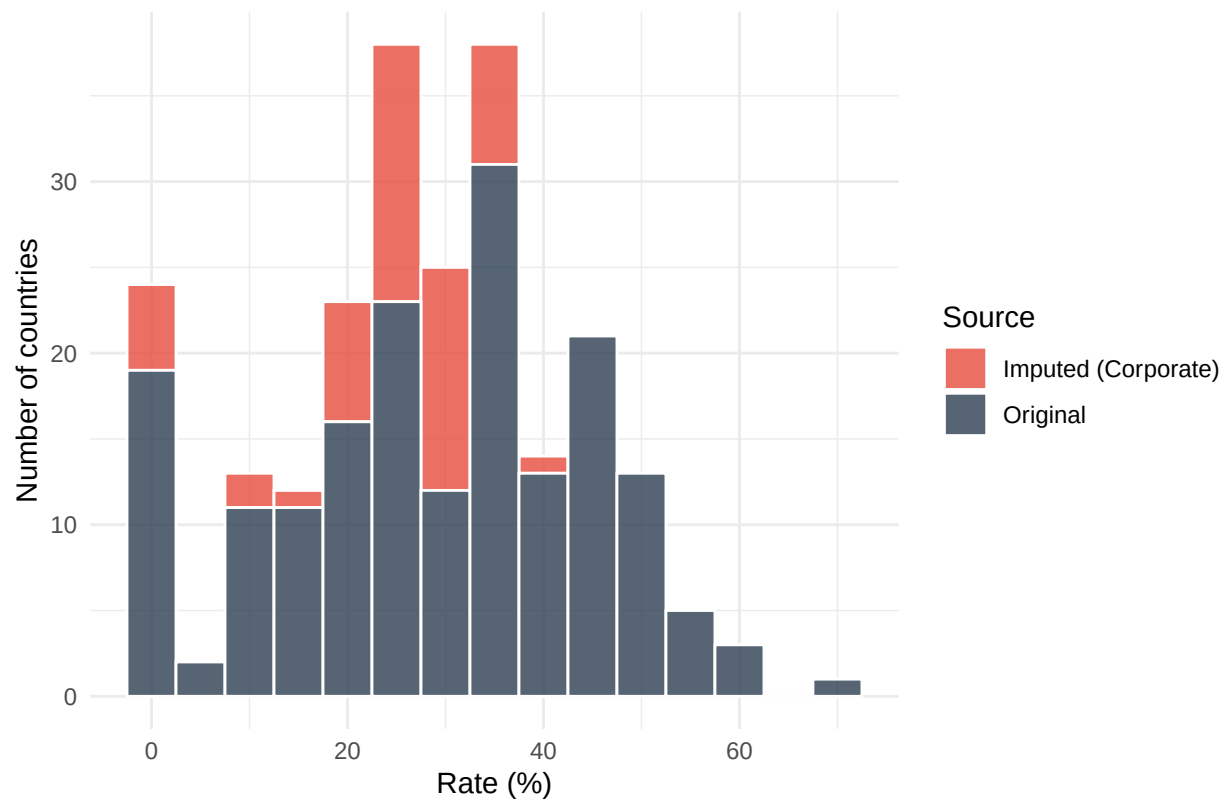


I add a visualizations of the corporate taxes imputed in income taxes for countires which don't have values. Always in terms of number of countries.

```
df_visu <- df_imputed %>%
  mutate(Source = ifelse(is.na(IncomeTaxTop), "Imputed (Corporate)", "Original"))

ggplot(df_visu, aes(x = IncomeTax_Imputed, fill = Source)) +
  geom_histogram(binwidth = 5, color = "white", alpha = 0.8) +
  scale_fill_manual(values = c("Original" = "#2c3e50", "Imputed (Corporate)" = "#e74c3c")) +
  labs(title = "Distinction of the distribution of tax rates (Original vs Imputed)",
       x = "Rate (%)", y = "Number of countries") +
  theme_minimal()
```

Distinction of the distribution of tax rates (Original vs Imputed)

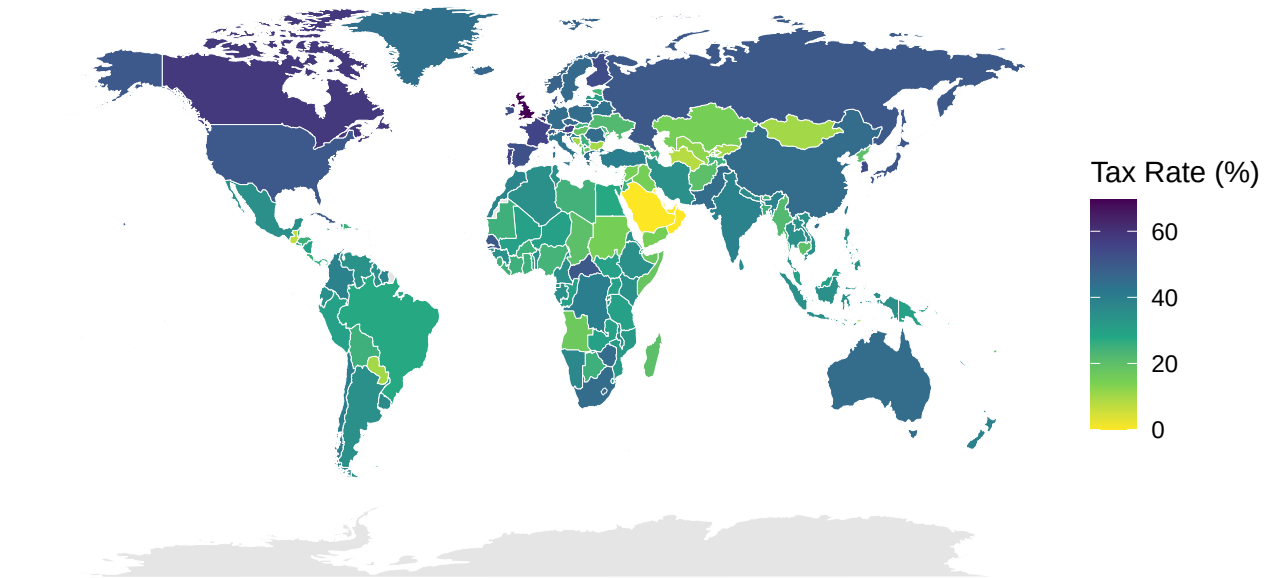


```
map_data_imputed <- left_join(world_map, df_imputed, by = "iso3")
```

```
ggplot(map_data_imputed, aes(long, lat, group = group, fill = IncomeTax_Imputed)) +
  geom_polygon(color = "white", size = 0.1) +
  scale_fill_viridis_c(option = "viridis", direction = -1, na.value = "gray90", name = "Tax Rate (%)") +
  labs(title = "Global Income Tax Rates with Imputed value",
       subtitle = "Missing income tax rates replaced by corporate tax rates") +
  theme_void() +
  coord_fixed(1.3)
```

Global Income Tax Rates with Imputed value

Missing income tax rates replaced by corporate tax rates



Third part

```
df_emploi <- read_delim("DD_EEC_ANNUEL_2024_data.csv", delim = ";", trim_ws = TRUE)
df_emploi
```

```
## # A tibble: 16,524 x 17
##   EEC_MEASURE UNIT_MEASURE PCS SEX AGE EMPFORM EMPSTA EDUC IMMI WKTIME
##   <chr>         <chr>      <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr>
## 1 EMP         _Z        21    F  Y20T24 _T    1_BIT _T    _T    _T
## 2 EMP         _Z        68    M  Y50T54 _T    1_BIT _T    _T    _T
## 3 EMP         _Z        65    M  Y30T34 _T    1_BIT _T    _T    _T
## 4 EMP         _Z        62    M  Y60T64 _T    1_BIT _T    _T    _T
## 5 EMP         _Z        33    M  Y40T44 _T    1_BIT _T    _T    _T
## 6 EMP         _Z        31    M  Y30T34 _T    1_BIT _T    _T    _T
## 7 EMP         _Z        65    _T  Y40T44 _T    1_BIT _T    _T    _T
## 8 EMP         _Z        52    _T  Y55T59 _T    1_BIT _T    _T    _T
## 9 EMP         _Z        48    _T  Y50T64 _T    1_BIT _T    _T    _T
## 10 EMP        _Z        33    _T  Y30T34 _T    1_BIT _T    _T    _T
## # i 16,514 more rows
## # i 7 more variables: ACTIVITY <chr>, UNDEREMP <chr>, UNEMPDUR <chr>,
## # COMPOHALO <chr>, OBS_STATUS <chr>, TIME_PERIOD <dbl>, OBS_VALUE <dbl>
```

```

df_emploi$UNIT_MEASURE <- NULL
df_emploi$EMPFORM <- NULL
df_emploi$EMPSTA <- NULL
df_emploi$UNDEREMP <- NULL
df_emploi$OBS_STATUS <- NULL
df_emploi$UNEMPDUR <- NULL
df_emploi$WKTIME <- NULL
df_emploi$ACTIVITY <- NULL
df_emploi$COMPOHALO <- NULL

df_plot <- df_emploi %>%
  filter(
    EEC_MEASURE == "UNEMPRATE",
    PCS == "_T",
    EDUC == "_T",
    IMMI == "_T",
    SEX != "_T",
    AGE %in% c("Y15T24", "Y25T49", "Y50T64")
  ) %>%
  mutate(
    Age_Label = case_when(
      AGE == "Y15T24" ~ "Youth from 15 to 24 years old",
      AGE == "Y25T49" ~ "Actives from 25 to 49 years old",
      AGE == "Y50T64" ~ "Seniors from 50 to 64 years old"
    ),
    Age_Label = factor(Age_Label, levels = c("Youth from 15 to 24 years old", "Actives from 25 to 49 years old", "Seniors from 50 to 64 years old")),
    Sex_Label = ifelse(SEX == "M", "Men", "Women")
  )

df_plot

```

```

## # A tibble: 6 x 10
##   EEC_MEASURE PCS   SEX   AGE   EDUC IMMI TIME_PERIOD OBS_VALUE Age_Label
##   <chr>      <chr> <chr> <chr> <chr> <chr>    <dbl>    <dbl> <fct>
## 1 UNEMPRATE _T    M    Y50T64 _T    _T      2024      5   Seniors from~
## 2 UNEMPRATE _T    M    Y25T49 _T    _T      2024     6.7 Actives from~
## 3 UNEMPRATE _T    F    Y25T49 _T    _T      2024     6.7 Actives from~
## 4 UNEMPRATE _T    F    Y50T64 _T    _T      2024      5   Seniors from~
## 5 UNEMPRATE _T    M    Y15T24 _T    _T      2024    19.1 Youth from 1~
## 6 UNEMPRATE _T    F    Y15T24 _T    _T      2024    18.4 Youth from 1~
## # i 1 more variable: Sex_Label <chr>

```

```

ggplot(df_plot, aes(x = Age_Label, y = OBS_VALUE, fill = Sex_Label)) +
  geom_col(position = "dodge", alpha = 0.9) +
  geom_text(aes(label = paste0(round(OBS_VALUE, 1), "%"),
    position = position_dodge(width = 0.9),
    vjust = -0.5, size = 3.5, fontface = "bold")) +
  scale_fill_manual(values = c("Men" = "#2c3e50", "Women" = "#e74c3c")) +
  labs(
    title = "Youth unemployment in 2024 by age and sex in France",
    x = NULL,
    y = "Unemployment rate (%)",
    fill = NULL,
  )

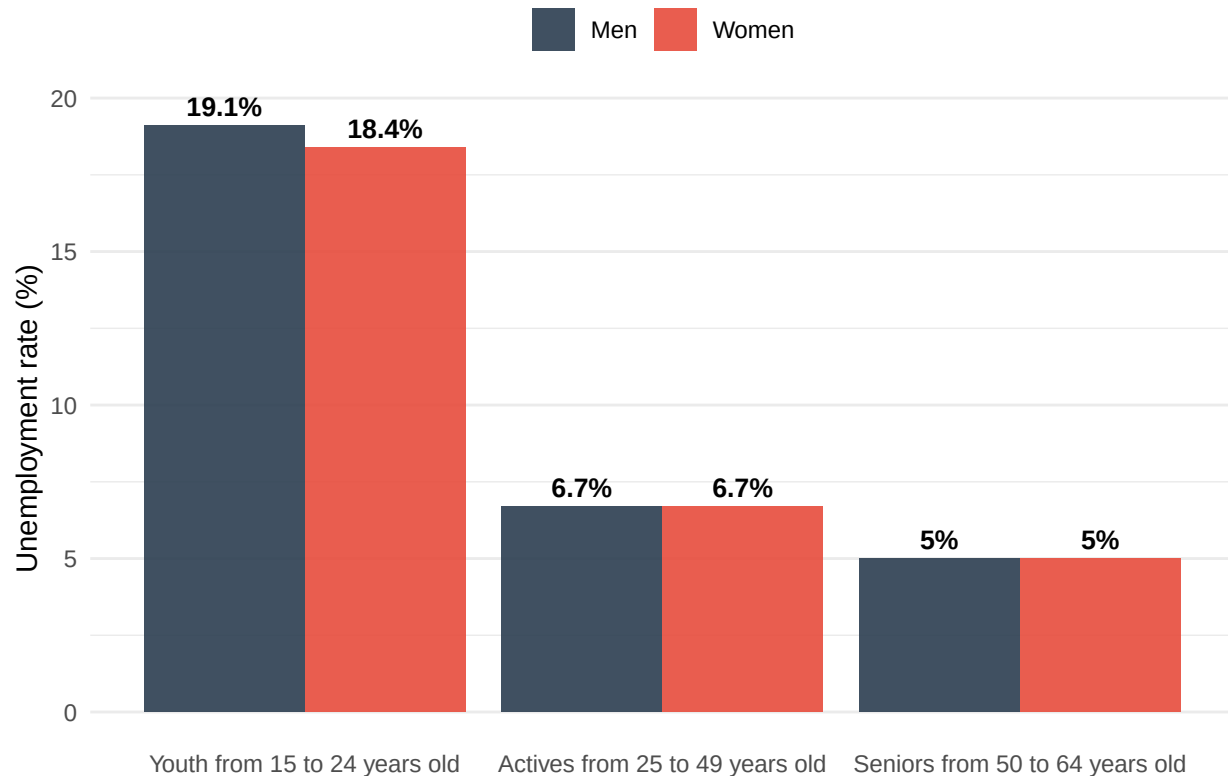
```

```

) +
theme_minimal() +
theme(
  legend.position = "top",
  panel.grid.major.x = element_blank(),
  plot.title = element_text(face = "bold", size = 14)
)

```

Youth unemployment in 2024 by age and sex in France



This visualization highlights an inequality in the French labor market which tell us that age is the primary determinant of unemployment risk more than gender. The story told us that: to start your career after your diploma, it's difficult to enter in. The youth category dominates the chart with unemployment rates nearing 20 percent, which is three times higher than the rates observed for actives workers or seniors.

This disparity highlights from school to work transition. It's often due to a period characterized by precarious contracts and friction between employment and unemployment. To illustrate this story effectively, a grouped bar chart was chosen to simultaneously compare age and gender, with age groups arranged chronologically. Contrary to what one might think, the difference between men and women is very close in terms of unemployment when you are young and equal when you are active or senior.